

CAMARENAFrancesca

Robotics | Software Engineer | Mechatronics Engineer

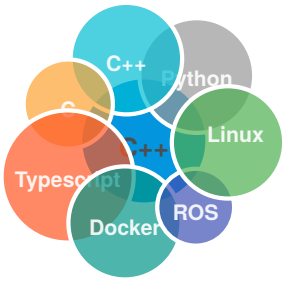
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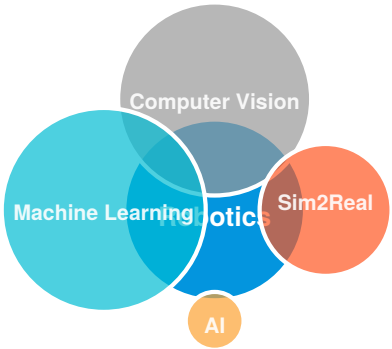
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Programming



Fields of Interest



Languages
English [bilingual]
★★★★★
French [proficient]
★★★★☆
Spanish [native]
★★★★★

Experience

Data Science & New Technologies Consultant

December 2023 – July 2024 Management Solutions, Lima, Peru

Digital transformation — retail banking automation

- Diagnosed inefficiencies in retail banking loan contract workflows (duplicity, non-traceability, inconsistent formatting) and designed TO-BE process models to address them.
- Led cross-functional workshops with legal, security and business stakeholders to validate process redesigns and drive alignment on proposed solutions.
- Created a standardized catalog of covenant clauses in coordination with legal teams.

Cloud migration — Cloudera to Azure Datalake

- Migrated critical data pipelines for Peru's largest banking entity from Cloudera to Azure, working hands-on with Spark, Hive, PySpark and SQL.
- Configured Bitbucket repositories, refactored legacy code to PySpark standards, and validated data integrity.
- Diagnosed and resolved pipeline incidents post-migration, applying structured debugging and data reconciliation at scale.

Information & Technology Intern

2022 United Nations – World Food Programme (WFP)

Process Automation · Power Automate · Power BI · SAP Requirements Analysis · Cross-team Collaboration

- Identified automation opportunities across CBT, HR and TEC divisions by conducting structured interviews with internal stakeholders.
- Designed and deployed automated workflows using Power Automate, Power BI and Power Query, reducing manual effort on routine operations and freeing capacity for higher-value work.
- Advised WFP programs on adopting new technologies for data recording, processing and storage – demonstrating the ability to translate technical solutions into organizational impact.
- Co-designed the CLTF vertical articulation project to strengthen local food supply chains, combining data-driven analysis with cross-program coordination across multiple stakeholders.

Application Engineer → Project Analyst

2020 – 2021 Andes Technology S.A.C.

Project Management · KUKA · Inventor · 3D Simulation · Client Engagement

- Led pre-engineering design for automation projects with clients, managing scope, milestones and stakeholder expectations from first contact to delivery.
- Designed automation systems and ran dynamic simulations of KUKA robots using Inventor and KUKA Sim, translating client requirements into viable technical proposals.
- Conducted on-site visits to collect requirements and identify pain points.
- Developed and delivered 3D printer training sessions for client personnel.

Industrial Engineering Trainee

2019 – 2020 Creditex S.A.

Python · PLC · Power BI · Machine Vision · Automation UI & Database Design · Project Planning

- Designed and built an automated fabric quality inspection system – selecting components, implementing M2M communication between PC and PLC via Python, and creating a defect-detection database.
- Built a user interface for image registration and quality statistics display, applying product and UX sensibility to an industrial context.
- Developed Power BI dashboards used by area supervisors to coordinate quality control operations, bridging technical data and operational decision making.
- Managed project milestones and a 2-person team using Microsoft Project, delivering on time within tight manufacturing constraints.

Selected Projects

2026 - Present **Modern Robot Learning** AI Lab - 42 AI

A modular robotics research platform for learning and experimentally evaluating modern approaches to robot perception, manipulation, imitation learning, reinforcement learning, and embodied foundation models.

`https://github.com/cuycorp/Modern-Robot-Learning`

2026 - Present **Duck - a thought tracker** Personal project

A local-first focus companion, desktop overlay application, that lets users quickly externalize distracting thoughts and return to their task.

`https://github.com/cuycorp/Duck`

2026 - Present **Tokenscope** Ecole 42

TokenScope is a collaborative platform that helps engineering teams understand where their LLM spending comes from by collecting usage traces, calculating estimated cost, visualizing live analytics, and generating grounded optimization recommendations.

`https://github.com/lynyam/tokenscope`

2025 **Minishell** Ecole 42

Developing an autonomous robot for underground exploration I was responsible for traversability analysis, navigation, exploration and some of the mapping tasks. The project received excellent reviews from the EU commission.

`https://github.com/cuycorp/minishell`

Honors & Awards

2013 **Honorable Mention, Model United Nations (MUN)** Harvard Model United Nations, Boston, USA

I was awarded the Honorable Mention at the committee of Islamic Educational, Scientific and Cultural Organization, under the topic of Women in Higher Education the oldest, largest, and most prestigious MUN conference, for high school students, in the world.

2018 **Half Scholarship, International Baccalaureate Diploma** United World College of India, Pune, India

I was chosen as one of 500 best applicants across Peru for the selective UWC network. The candidates between the ages of 16 and 19 were selected, for a program dedicated to turn young people's idealism and energy into: lifelong action, responsibility, and empathy.

Education

2024 - now **MSc. Computer Software Engineering** Ecole 42, France

A project-based and peer-to-peer computer science program ranked the world's 3rd most innovative university in the 2025 and 2026 WURI rankings. Training focused on system-level programming, C, Linux, Git, and shell development, with extensive team projects in an autonomous, collaborative learning environment.

2017 - 2022 **BSc. Mechatronics Engineering** Pontifical Catholic University of Peru, Lima, Peru

The Pontifical Catholic University of Peru (PUCP) is ranked No. 1 in Peru and No. 10 in Latin America. My studies focused on mechanical and electronic systems and the design of integrated hardware–software automation systems. Thesis: design of an EEG-controlled omnidirectional wheelchair.

2015 - 2016 **Bilingual International Baccalaureate Diploma** Newton College, Lima, Peru

Top bilingual Peruvian-British school, focused on developing inquisitive, outspoken and well-rounded individuals that have intercultural understanding.

October 5, 2026

CAMARENA PUCUHUANCA, Monica Francesca